

The Medium — Action Plan: Open Problems and Priorities

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1 Overview

This document replaces the original Smolin/Rovelli/Hossenfelder critical review action plan. Many issues from that review have been addressed in the current document revisions. This plan tracks what remains open, what has been resolved, and the priority order for future work.

1.1 What Has Been Resolved

The following issues from the original review have been addressed in the current document set (docs 0–5):

- **SU(3) claim corrected** — body and abstract now correctly state $U(1) \times U(2)$ as the established group; SU(3) moved to open derivation status (Doc 3)
- **LQG minimum area match removed** — 4π discrepancy between The Medium and LQG formulas now explicitly noted (Doc 4)
- **Diffeomorphism invariance claim hedged** — "inherits" replaced with structural motivation shared; constraint algebra closing identified as open (Doc 4)
- **Proton mass labeling fixed** — "Exact" changed to "Exact (by construction)" in all tables (Docs 1, 2, 4, 5)
- **G and m_p variation predictions flagged** — moved from predbox to openbox; explicitly noted as appearing observationally ruled out with specific numbers (Doc 5)
- **H₀ predbox softened** — "Prediction" changed to "Interpretation"; selected value vs. derived value distinction made clear (Doc 5)
- **$\Omega_\Lambda = 2/3$ relabeled** — "Prediction" changed to "Consequence of G formula"; not independently testable (Doc 5)
- **Λ hierarchy claim made conditional** — floor derivation identified as required; "dissolving" language removed (Docs 3, 5)
- **ADM summary corrected** — "recovered" replaced with schematic structure present; missing components listed (Doc 4)
- **Pion added to hadron table** — explicitly noted as not explained by current formula (Doc 2)
- **"Complete chain" removed** — changed to "connected chain. . . with open calculations explicitly identified" (Doc 2)
- **Spin identification qualified** — SU(2) representation theory noted as not yet derived (Doc 4)

- **Color charge claim qualified** — "are color charge" changed to "structurally analogous to color charge labels" (Doc 2)

2 FATAL Issues

2.1 FATAL [F1] Foundation connection — substantially addressed, four gaps remain

Doc 1 has been substantially revised. The following have been resolved:

- ☒ M defined: M is the substrate itself. Not a geometric space.
- ☒ ι is a distinction within M , not an event (no sequence implied).
- ☒ $\phi : M \rightarrow \{+1, -1\}$ replaced by $c : D \times D \rightarrow \{\text{same, different}\}$. Character is relational, not intrinsic.
- ☒ Relations between co-existing distinctions are unavoidable (self-grounding argument: follows from singularity of M , same logic as distinction itself requiring no cause).
- ☒ Adjacency constraint reframed as informational, not topological.
- ☒ Closure condition identified as the localization mechanism: closed structures are self-contained; open chains are unstable and must close.
- ☒ Degree-2 argument stated and its implication documented (see below).

Four gaps remain open:

2.1.1 Gap 1 — Degree-2 not yet proven

The argument that each ι distinction has relational degree exactly 2 (from the two sides of its split) is plausible but not rigorous. Nothing in the current formulation rules out higher-degree nodes. If degree is not fixed at 2, the triangle is not uniquely minimal and the localization argument weakens.

Testable implication: If degree 2 is correct, K_3 is the only primitive closed structure. All higher configurations (K_4 , K_5) must be composite. This is consistent with QCD but must be verified against the full particle spectrum.

Action: Derive degree-2 from the binary split rigorously, or identify the additional structure that allows higher degree.

2.1.2 Gap 2 — Mathematical construction of N from c not specified

N_{ij} is described as "the coherence of comparison across all paths through G ." The precise construction — how the binary comparison $c(i, j) \in \{\text{same, different}\}$ gives rise to the continuous real field $N_{ij} \in \mathbb{R}$ — is not written down. The candidate identification $N_{ij} \sim (L^{-1})_{ij}$ is plausible but circular (L depends on G , G depends on N).

Action: Write an explicit non-circular construction of N from c .

2.1.3 Gap 3 — Complex N has no source

The full Lagrangian and any route to $SU(3)$ symmetry requires $N \in \mathbb{C}$. Neither the binary comparison c nor the real-valued coherence argument gives a natural origin for complex phases. Promoting c to a $U(1)$ -valued comparison (phases rather than binary values) is the most natural candidate but has not been developed.

Action: Identify the mechanism that produces complex phase in N .

2.1.4 Gap 4 — Frustrated triangle ground state not worked out

The triangle K_3 necessarily has at least one same-comparing edge (it cannot be 2-colored without adjacent same-color nodes). The adjacency constraint says same-comparing negotiations are informationally empty. What is N on the frustrated edge? If $N = 0$, the triangle loses an edge and the Laplacian changes. If $N \neq 0$, the constraint is violated. The degenerate ground state of the frustrated triangle — from which spin is supposed to emerge — has not been derived.

Action: Work out the ground state of the frustrated triangle under the informational adjacency constraint. Determine whether frustration degeneracy corresponds to the double degeneracy of $\lambda = 3$.

2.2 RESOLVED [F2] "From First Principles" on Doc 2 title page

☒ Subtitle changed to "Structure from Substrate Primitives; Scale from Observation."

The distinction: structural results (triangle topology, $\lambda = 3$, three-fold mass ratio, quantization from closure, confinement from self-containment) are genuinely derived from the substrate primitives. The numerical scale (ε , H_0 , γ) requires empirical input. The new subtitle reflects this accurately.

3 SEVERE Issues

3.1 SEVERE [S1] G and m_p variation rates appear observationally ruled out

The cosmological evolution equations give:

- $\dot{G}/G \approx -1.5 \times 10^{-10}/\text{yr}$
- $\dot{m}_p/m_p \approx 7 \times 10^{-11}/\text{yr}$

Current observational bounds are approximately $200\times$ tighter for G and approximately $7000\times$ tighter for m_p . These are now flagged in Doc 5 as "may be observationally ruled out" in an openbox. This is correct but incomplete — the underlying equations need to be examined.

3.1.1 Action steps

1. ☐ Determine whether a physical mechanism exists that suppresses local variation while allowing secular variation over Hubble timescales. This is the only route to saving these predictions. Candidates: the floor condition at ρ_Λ , or a relational definition of G and m_p that makes them observer-dependent rather than absolute.

2. ☐ If no such mechanism can be identified, remove the evolution equations for G and m_p from the description entirely, or relocate them to a section labeled "candidate equations pending constraint analysis."
3. ☐ Λ variation ($d\Lambda/dt = -2\Lambda H_0$) has no current constraint but follows from the same equations. Its status is linked to items 1 and 2.

3.2 SEVERE [S2] Fermionic statistics not produced by the bosonic Lagrangian

The negotiation Lagrangian is bosonic. Canonical quantization gives $[\hat{a}, \hat{a}^\dagger] = 1$ (Bose-Einstein statistics). The proton is a spin-1/2 fermion. This is identified as a primary open problem in Docs 3 and 4 but no progress has been made on either candidate resolution.

3.2.1 Action steps

1. ☐ Investigate composite antisymmetrization first: show whether three bosonic quanta in the antisymmetric combination on the triangle nodes acquire fermionic exchange behavior under interchange of two protons. Write out the explicit state and check the exchange statistics.
2. ☐ If antisymmetrization fails: investigate Grassmann-valued N and rederive the Lagrangian with anticommuting fields.
3. ☐ Document whichever path is pursued, even if it fails, so a mathematician reading this knows what has been tried.

3.3 SEVERE [S3] The energy formula $E(n,k) = \epsilon \cdot k \cdot n$ is not fully derived

The formula is presented as following from the Hamiltonian eigenvalue equation, and for complete graphs K_n with mode $k = 1$ (ground non-trivial mode), $E = \epsilon \cdot n$ follows directly from $\lambda = n$. But the k factor for higher modes is introduced without derivation, and the table in Doc 4 uses $E_n = \epsilon \cdot n$ without specifying k . For the rho meson ($n=2$) the predicted value of 625.6 MeV already differs from observed (775 MeV) by 19%. The formula's domain of validity is unclear.

3.3.1 Action steps

1. ☐ Derive explicitly what k represents for higher negotiation modes on complete graphs and write out $E(n,k)$ with k specified for each entry in the hadron table.
2. ☐ Address the rho meson 19% discrepancy: is the formula valid for $n=2$ or does the two-node topology require a different treatment?

4 SERIOUS Issues

4.1 SERIOUS [SR1] The symmetry group of the triangle Lagrangian not established

Doc 3 now correctly states that the group is $U(1) \times U(2)$, not $SU(3)$, and identifies $SU(3)$ as an open derivation. But the path from $U(1) \times U(2)$ to $SU(3)$ — if it exists — has not been investigated. This blocks the color charge identification and the strong force correspondence.

4.1.1 Action steps

1. ☐ Compute the symmetry group of $N^\dagger L N$ for K_3 with complex N explicitly. Confirm $U(1) \times U(2)$.
2. ☐ Identify what additional invariant — a cubic ε_{ijk} term, a different action on edge states — could promote the symmetry to $SU(3)$.
3. ☐ If $SU(3)$ cannot be reached: accept $U(1) \times U(2)$ as the established group and demote all color charge correspondence to "structural analogy at the level of node counting."

4.2 SERIOUS [SR2] The pion is absent from the hadron spectrum

The pion (135 MeV) is the lightest hadron and the most important test case. It is now explicitly listed in the Doc 2 hadron table as "not explained by this formula." A physicist reviewing this will treat the pion's absence as a primary failure of the spectrum description, not a minor gap. It requires a qualitatively different physical mechanism (chiral symmetry breaking) that is not present in the current description.

4.2.1 Action steps

1. ☐ Investigate whether the pion can be accommodated as a two-node graph with a non-complete topology (i.e., a single edge rather than the complete graph), or as a Goldstone mode of a broken symmetry in the negotiation field.
2. ☐ If neither works: explicitly state in Doc 2 that the light meson sector requires a mechanism outside the current description, and identify what that mechanism would need to be.

4.3 SERIOUS [SR3] The problem of time is unaddressed

The Lagrangian and Hamiltonian use $\partial N / \partial t$. In a pre-geometric description, t must emerge from the substrate. The description currently uses background time without acknowledging this as a structural gap. This is particularly acute given the LQG analogy in Doc 4 — the problem of time is one of LQG's central unresolved issues.

4.3.1 Action steps

1. ☐ Add a brief section to Doc 1 or Doc 4 titled "The Problem of Time" identifying what t is in the description. Candidates: a discrete count of ι negotiation steps; a relational time defined by the ι cycle count relative to a reference mode; or a classical background retained as an approximation. State which is assumed and why.

4.4 SERIOUS [SR4] The Hamiltonian floor condition is underived

Several results are conditional on the Hamiltonian being undefined below the vacuum floor ρ_Λ . This is now correctly flagged as conditional in Docs 3 and 5. But the floor condition is also the only route to addressing the G and m_p variation tensions (SEVERE [S1]) and the only route to the Λ hierarchy resolution. Its derivation is therefore load-bearing for multiple claims.

4.4.1 Action steps

1. ☐ Attempt to derive the floor condition from the substrate Lagrangian: show that modes below ρ_Λ have zero amplitude in the path integral, or that the Hamiltonian is ill-defined for configurations below the Hubble-floor energy density.
2. ☐ If derivation is not achievable: consolidate all conditional results into a single "conditional on floor derivation" section in Doc 1, so the dependency is visible in one place.

5 MODERATE Issues

5.1 MODERATE [M1] The LQG minimum area discrepancy has no resolution

The 4π factor difference between The Medium's area formula and the LQG formula is now honestly flagged. But its origin is unknown. A sphere-area normalization or solid-angle factor is suggested as a candidate; this has not been checked.

5.1.1 Action steps

1. ☐ Work out the iota surface area calculation explicitly and determine where a factor of 4π could enter. Either find it or confirm it is absent and the correspondence is structural only.

5.2 MODERATE [M2] Metric identification $h_{ij} (L^{-1}) (L(L^{-1}))_{ij}$ not justified

The inverse Laplacian is the graph Green's function, encoding signal propagation, not a metric in the Riemannian sense. The identification requires showing correct transformation under graph automorphisms, correct signature, and recovery of flat space in vacuum.

5.2.1 Action steps

1. ☐ Either derive the metric properties of $(L^{-1})_{ij}$ explicitly, or replace the identification with: "The inverse Laplacian is a candidate for the emergent metric; establishing its Riemannian properties is open work."

5.3 MODERATE [M3] Barbero-Immirzi derivation remains the single most valuable calculation

$\gamma \approx 0.2375$ appears as the residual correction in three separate gaps: $G(3-6)\backslash\gamma$ is fixed by black hole entropy counting. If the same entropy counting performed on the negotiation graph gives $\gamma \approx 0.2375$, it would simultaneously close all three gaps and confirm that γ is not a free parameter. This is the single calculation with the highest payoff across the entire description.

5.3.1 Action steps

1. ☐ Attempt the Bekenstein-Hawking entropy count on the negotiation graph of a black hole horizon: count the microstate degeneracy of the negotiation modes on the horizon graph and require it to equal $A/4l_{\text{Planck}}^2$.

2. $\square$ Even a partial result — showing the entropy count has the right structure to fix γ — would be significant.

6 Priority Order for Next Working Session

In order of what unblocks the most downstream work:

1. **FATAL-F2 (text fix):** Remove "from First Principles" from Doc 2 title page. Takes five minutes. Removes an immediate credibility problem.
2. **FATAL-F1 (foundational gap):** Connect ϕ to N , or consolidate around N as the true primitive. This is the deepest structural gap. Even a clear statement of what N is and where it comes from would help.
3. **MODERATE-M3 (Barbero-Immirzi):** The entropy counting calculation. High payoff: closes three gaps simultaneously and may constrain the G and m_p variation rates if the floor condition falls out of it.
4. **SEVERE-S1 (G and m_p variation):** Determine whether a suppression mechanism exists. If not, formally withdraw those predictions. Leaving them as "may be ruled out" is better than before but a physicist will want resolution.
5. **SERIOUS-SR1 (symmetry group):** Complete the group theory calculation for K_3 . This either establishes the $SU(3)$ correspondence or closes the door on it, which is better than the current uncertainty.
6. **SERIOUS-SR2 (pion):** Investigate Goldstone / non-complete-graph accommodations. If neither works, state so explicitly. The pion cannot be ignored indefinitely.
7. **SEVERE-S2 (fermionic statistics):** Try the composite antisymmetrization route. If it works it resolves the most fundamental physical gap in the description.
8. **SERIOUS-SR3 and SR4 (time and floor condition):** Add the problem of time section; attempt the floor derivation. Both are required for the description to be internally consistent at a foundational level.

7 What Does Not Need Fixing

- The internal consistency triangle (G , H_0 , m_{proton}) is a genuine structural observation and should be highlighted, not hedged further.
- The openbox and plainbox structure across Docs 2–5 is good practice and should be extended to any new sections rather than removed.
- The honest labeling of open calculations, failed predictions, and structural candidates rather than established results is the right register for the current state of the work. Do not retreat from this.
- Doc 0 (personal introduction) is in good shape and sets the right tone.
- The topology hierarchy ($n = 2, 3, 4, 5$ complete graphs) is a genuine structural result even if incomplete. Keep it.